

Automated Medical Image Boundary Detection

- Course: Computer Vision Techniques
- Name: Sarathkumar Venkatesan
- Instructor: Dr Meenakshi
- Institution: SRM Institute of Science & Technology

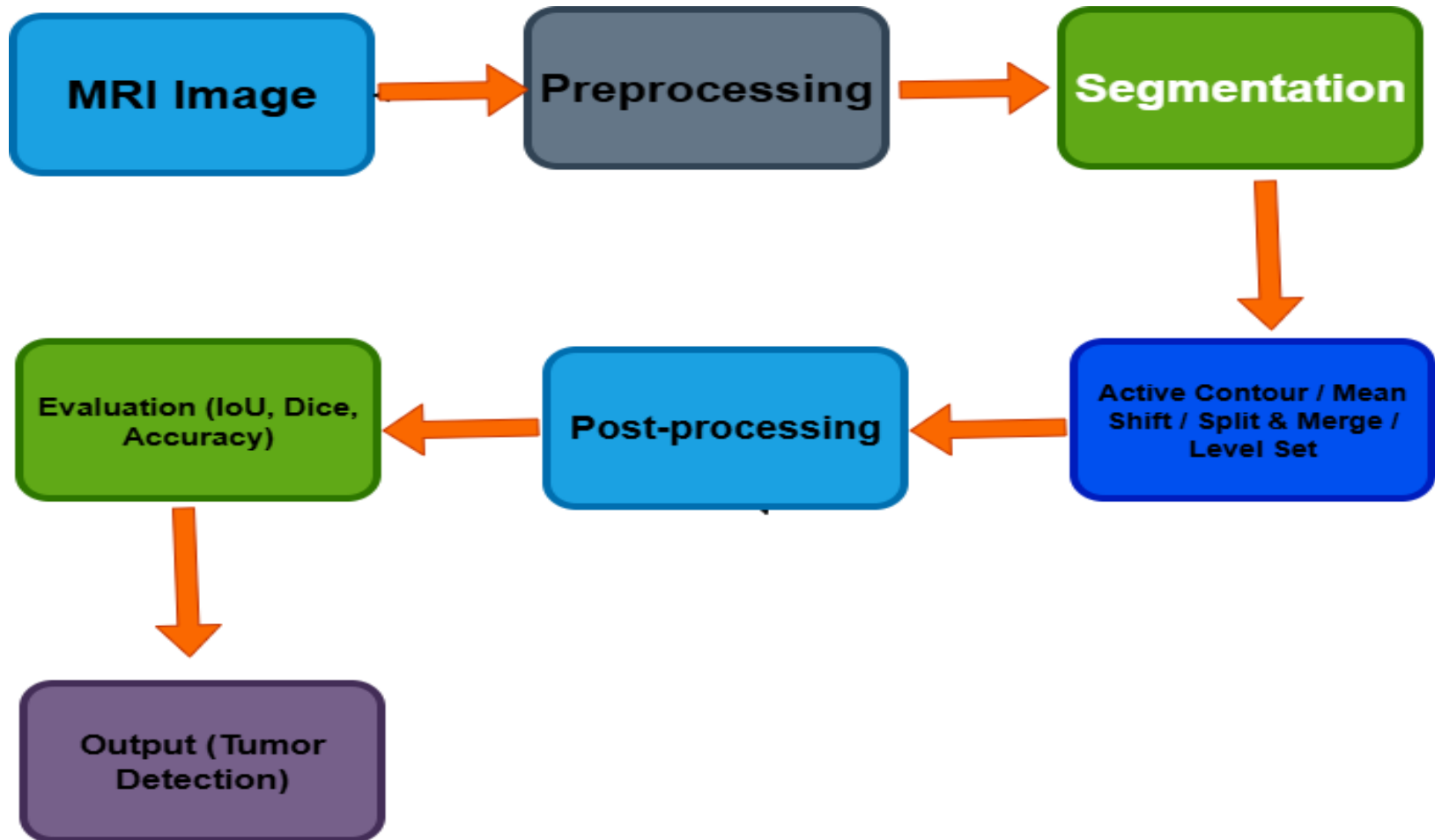
Problem Statement

A hospital aims to assist doctors in accurately identifying tumor boundaries in MRI images. The objective is to develop an automated system that can detect object boundaries and highlight regions of interest, enabling faster and more reliable medical diagnosis.

Objectives

- Develop an automated system to detect tumor boundaries in MRI images and highlight regions of interest.
- Active Contours, Level Set, Mean Shift, and Split & Merge.
- Perform boundary detection visualisation to identify abnormal regions clearly.
- Compare segmentation methods and evaluate their accuracy using sample MRI images with performance metrics

System Overview



Methodology

- Image acquisition & preprocessing
- Apply multiple segmentation methods
- Generate binary masks
- Evaluate using IoU, Dice, and accuracy
- Compare results to find the best method

Techniques Used

- Active Contour (Snakes): Detects object boundaries by evolving a curve towards the tumor edges.
- Level Set Method: Captures complex and irregular tumor shapes using contour evolution.
- Mean Shift Segmentation: Groups pixels based on intensity similarity to identify regions.
- Split and Merge Segmentation: Divides the image into regions and merges similar areas based on homogeneity.

Algorithm Details

- **Preprocessing:** Apply smoothing (Gaussian filter) and dynamic range normalization to improve image quality.
- **Active Contour (Snakes):** Initializes a contour and iteratively moves it to fit the tumor boundary.
- **Level Set Method:** Evolves contours to detect complex and irregular shapes.
- **Mean Shift Segmentation:** Clusters pixels based on intensity (image resized for efficient processing).
- **Split & Merge:** Divides the image into regions and merges similar areas based on uniformity.

Implementation

```
# Active Contour
# Smooth image
image_smooth = gaussian(image, sigma=2)

# Initial contour (circle)
h, w = image.shape

radius = min(h, w) // 7 # increase range

s = np.linspace(0, 2*np.pi, 400)
r = h//2 + radius * np.sin(s)
c = w//2 + radius * np.cos(s)

init = np.array([r, c]).T

# Apply snake
snake = active_contour(
    image_smooth,
    init,
    alpha=0.01,
    beta=2,
    gamma=0.005
)
```

```
# Split and Merge Segmentation
def split_and_merge(img, threshold=0.001):
    h, w = img.shape
    output = np.zeros_like(img)

    def split(x, y, size):
        region = img[x:x+size, y:y+size]

        if np.var(region) < threshold or size < 8:
            output[x:x+size, y:y+size] = np.mean(region)
        else:
            half = size // 2
            split(x, y, half)
            split(x+half, y, half)
            split(x, y+half, half)
            split(x+half, y+half, half)

    split(0, 0, min(h, w))
    return output

split_merge_result = split_and_merge(image)
```

```
# Mean Shift Segmentation
# Convert image to 2D feature space
# Resize image to speed up
image_small = cv2.resize(image, (128, 128))

flat_image = image_small.reshape(-1, 1)

# Apply Mean Shift
ms = MeanShift(bandwidth=0.1)
ms.fit(flat_image)

labels = ms.labels_
segmented = labels.reshape(image_small.shape)

show_image("Mean Shift Segmentation", segmented)
```

```
# Level Set
# We simulate level set using contour detection

# Convert to binary
_, thresh = cv2.threshold((image*255).astype(np.uint8), 127, 255, cv2.THRESH_BINARY)

# Find contours
contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

# Draw contours
level_set_img = cv2.cvtColor((image*255).astype(np.uint8), cv2.COLOR_GRAY2BGR)
cv2.drawContours(level_set_img, contours, -1, (0,255,0), 2)

show_image("Level Set (Contours)", level_set_img, cmap=None)
```


Results

```
Ground truth loaded
```

```
IoU Scores:
```

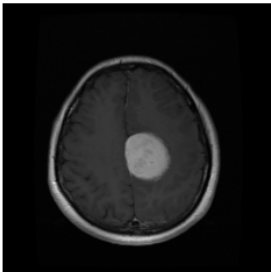
```
Active Contour: 0.9183
```

```
Mean Shift: 0.4588
```

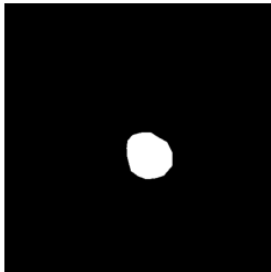
```
Split & Merge: 0.0697
```

```
Level Set: 0.4326
```

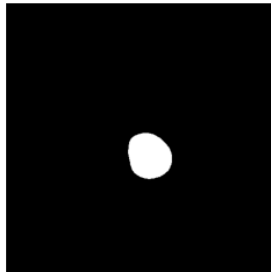
Original



Ground Truth



Active Contour



Mean Shift



Split & Merge



Level Set



Performance Analysis

	Method	IoU	Dice	Accuracy	Precision	Recall	Time (sec)
0	Active Contour	0.918331	0.957427	0.998081	0.993326	0.924032	0.001259
1	Mean Shift	0.458762	0.628975	0.976944	0.503786	0.836955	0.004375
2	Split & Merge	0.069684	0.130288	0.688267	0.069684	1.000000	0.003377
3	Level Set	0.432570	0.603908	0.977032	0.505507	0.749877	0.000933



BEST METHOD:

Active Contour

IoU: 0.9183308978730313

Advantages

- Automated detection
- Improved accuracy
- Better visualization
- Comparative analysis
- Efficient processing
- Supports medical diagnosis

Limitations

- Sensitive to noise and contrast
- Requires parameter tuning
- Computationally expensive (Mean Shift)
- Initialization issues (Snakes)
- May struggle with complex boundaries

Conclusion

- Developed an automated tumor detection system
- Compared multiple segmentation methods
- Evaluated using standard metrics
- Identified best performing technique

Thank you